

Trigonometric Substitutions and Inverse Trigonometric Expressions

Recovering algebraic expressions in x

For every problem, express the requested quantity entirely in terms of x . Use the stated substitution or introduce an angle such as $\theta = \arcsin u$, simplify radicals, choose signs from the relevant range, and state the domain when it is not already stated.

Reference sheet

$$\sin^2 \theta + \cos^2 \theta = 1, \quad 1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta.$$

$$\sin(2\theta) = 2 \sin \theta \cos \theta, \quad \cos(2\theta) = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1, \quad \tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$$

Principal ranges used below are

$$\arcsin u \in [-\pi/2, \pi/2], \quad \arccos u \in [0, \pi], \quad \arctan u \in (-\pi/2, \pi/2).$$

For arcsec problems with positive argument restrictions, use $\theta \in [0, \pi/2)$. For unrestricted arcsec, use $\theta \in [0, \pi]$ with $\sin \theta \geq 0$. For arccsc, use $\theta \in [-\pi/2, \pi/2]$ with $\cos \theta \geq 0$, and for arccot, use $\theta \in (0, \pi)$.

1 Direct Substitution Recovery

Use the stated substitution and angle range. These are the conversions that often occur at the final stage of a trigonometric-substitution integral.

1. $x = 3 \sin \theta$, where $-\pi/2 \leq \theta \leq \pi/2$. Express $2 \cos \theta$ in terms of x .

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2. $x = 5 \tan \theta$, where $-\pi/2 < \theta < \pi/2$. Express $\frac{1}{2} \sin \theta$ in terms of x .

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3. $x = 2 \sec \theta$, where $0 \leq \theta < \pi/2$. Express $\frac{1}{4} \sin \theta$ in terms of x .

4. $x = 4 \sin \theta$, where $-\pi/2 < \theta < \pi/2$. Express $\tan \theta$ in terms of x .

5. $x = 3 \tan \theta$, where $-\pi/2 < \theta < \pi/2$. Express $\sec \theta$ in terms of x .

6. $x = 6 \sec \theta$, where $0 \leq \theta < \pi/2$. Express $\tan \theta$ in terms of x .

7. $x = 2 \sin \theta$, where $0 < \theta \leq \pi/2$. Express $3 \csc \theta$ in terms of x .

8. $x = 7 \tan \theta$, where $-\pi/2 < \theta < \pi/2$. Express $4 \cos \theta$ in terms of x .

9. $x = 5 \sec \theta$, where $0 < \theta < \pi/2$. Express $2 \cot \theta$ in terms of x .

2 Reciprocal Compositions

Start by setting θ equal to the inverse trigonometric expression. These problems require little more than taking a reciprocal, but domains still matter.

10. $5 \csc\left(\arcsin \frac{x}{5}\right)$

11. $\sec(\arccos 3x)$

12. $2 \cot(\arctan 2x)$

13. $\cos\left(\operatorname{arcsec} \frac{x}{4}\right)$

14. $\tan(\operatorname{arccot} 5x)$

3 One Pythagorean Step

Use one Pythagorean identity, then choose the sign from the range of θ .

15. $3 \cos\left(\arcsin \frac{x}{3}\right)$

16. $\sin(\arccos 4x)$

17. $\tan\left(\arcsin \frac{x}{2}\right)$

18. $5 \sec\left(\arctan \frac{x}{5}\right)$

19. $\cos(\arctan 4x)$

20. $\cot\left(\arccos \frac{x}{6}\right)$

21. $\csc(\arccos 2x)$

22. Simplify $3 \cos(\arcsin(x/4))$.

23. Simplify $2 \sin(\arctan(3x))$.

24. For $0 < x \leq \frac{1}{2}$, simplify $\frac{1}{5} \tan(\arccos(2x))$.

25. Simplify $\sec(\arcsin(2x))$.

4 Sign-Sensitive Compositions

These are the cases where the principal range can force an absolute value or a sign factor.

26. $\sin\left(\operatorname{arcsec} \frac{x}{3}\right)$

27. $\tan\left(\operatorname{arcsec} \frac{x}{2}\right)$

28. $\cot\left(\operatorname{arccsc} \frac{x}{5}\right)$

29. $\cos(\operatorname{arccsc} 3x)$

30. $\sec\left(\operatorname{arccsc} \frac{x}{2}\right)$

31. $\csc(\operatorname{arcsec} 4x)$

32. $\sin\left(\operatorname{arccot} \frac{x}{4}\right)$

33. $\cos\left(\operatorname{arccot} \frac{x}{4}\right)$

5 Double-Angle Compositions

First recover the inner trigonometric quantities, then apply a double-angle identity.

34. $\sin\left(2 \arcsin \frac{x}{3}\right)$

35. $\cos(2 \arctan 2x)$

36. $\sin\left(2 \arctan \frac{x}{5}\right)$

37. $\tan\left(2 \arctan \frac{x}{3}\right)$

38. Simplify $2 \sin(2 \arctan x)$.

39. Simplify $\cos(2 \arcsin(x/3))$.

40. Simplify $\frac{1}{2} \tan(2 \arctan(x/2))$.

6 Final Mixed Review

Use whichever method is appropriate. The section title no longer tells which identity or sign issue is involved.

41. For $x \geq 3$, simplify $4 \sin(\operatorname{arcsec}(x/3))$.

42. For $x \neq 0$, simplify $\cot(\arctan(x/2))$.

43. $\cos\left(2 \operatorname{arcsec} \frac{x}{2}\right)$

44. $\sin\left(2 \operatorname{arcsec} \frac{x}{4}\right)$

45. For $x \geq 2$, simplify $3 \cos(2 \operatorname{arcsec}(x/2))$.

46. Simplify $\sin(2 \arccos x)$.

Final check

For each answer, ask:

1. Is the answer entirely in terms of x ?
2. Did I use the correct Pythagorean, reciprocal, or double-angle identity?
3. Did I choose the sign from the inverse function's principal range or the stated range of θ ?
4. Does the final expression have the same domain as the original one?